

can adjust the counter time setting based on their experience, providing control over the cooking process. This way a user has some control over the cooking time. These factors ensure that millets are properly cooked under all conditions.

Circuit and working

The circuit diagram of millet cooker is divided into two sections. The first comprises the power supply, counter, and MOSFET sections and the second the comparator and alarm sections. Fig. 2 shows the power supply, counter, and MOSFET sections. The input terminals of the AC-DC converter are connected to 230V AC through on/off switch S1. The +Vdc output is connected to H1 terminal of the heater coil. H2 terminal is connected to the drain of IRF540. LED1 is connected to +Vdc through resistor R1, to indicate that the power is on. The input terminal of DC-DC converter module XL7015 is connected to +Vdc through D1 and R2. Filter capacitor C1 helps to overcome very short disruptions in the 230V AC power supply. DC-DC converter output voltage is adjusted to get +5V power supply.

The 14-bit binary counter (IC1) generates the user-controlled time period. It has a built-in oscillator with adjustable frequency that

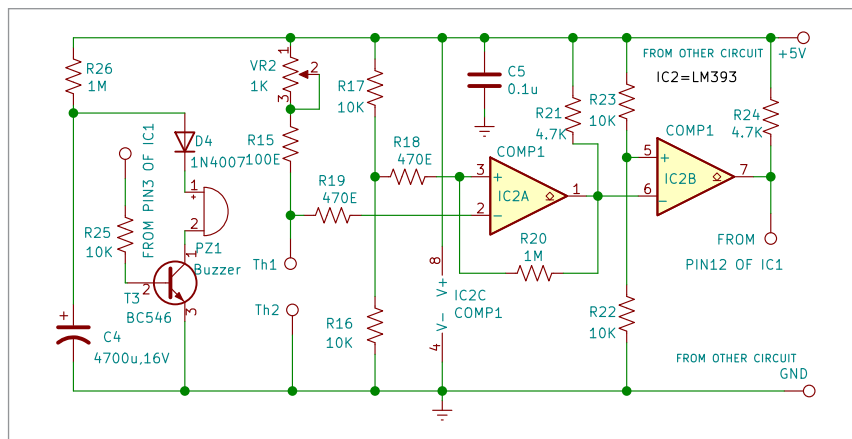


Fig. 4: Circuit diagram of the comparator and alarm sections

is achieved through resistors and potentiometers. Its pins 9, 10, and 11 are used to decide the oscillator frequency. Timing capacitor C2 is connected to pin 9. Pot VR1 is connected through series resistor R7 to pin 10. Resistors R4, R5, and R6 are connected to pin 11. Reset pin 12 is connected to the comparator output (pin 7 of IC2), as shown in Fig. 4.

When RST pin 12 goes low, the counter starts counting. Pin 3 (which is 14th stage output Q13) of counter IC1 is initially low. When the counter overflows, pin 3 goes high. To pin 7 (4th stage output Q3), red LED3 is connected through R8. This pin toggles and LED3 keeps blinking, indicating that the counter is counting. Fig. 3 shows the oscillator waveforms at pin 11 (yellow) and at pin 7 (blue). The measured time periods generated by the counter are as follows.

When VR1 setting is minimum, oscillator frequency will be around 7.35Hz. This means that the time period will be around 19 minutes and 12 seconds.

When VR1 setting is maximum, the oscillator frequency will be around 2.53Hz. This means the time period will be around 53 minutes and 41 seconds.

Fig. 4 shows the circuit diagram of the comparator and alarm sections. It uses dual comparator LM393 (IC2). To the non-inv pin 3 of COMP1, 2.5V is applied through the potential divider formed by resistors R16 and R17. To the inv pin 2, thermistor (10k NTC) is connected through terminals Th1 and Th2. The thermistor is powered through resistor R15 and potmeter VR2.

When the power is switched on, the voltage at pin 2 is greater than 2.5V because the thermistor is at room temperature (about 10k). COMP1 output at pin 1 is low.

COMP2 is configured as an inverting stage. Hence, output at its pin 7 is high. As the heater temperature goes on increasing, the resistance of the thermistor goes on reducing exponentially. When resistance goes below R15 plus VR2, the COMP1 output goes high and COMP2 output goes low. Pin 7 of COMP2 is connected to pin 12 (RST) of IC1, so the counter starts counting. Resistor R20 introduces hysteresis for preventing undesired switching.

Power-on sequence. Before turning on switch S1, ensure that the

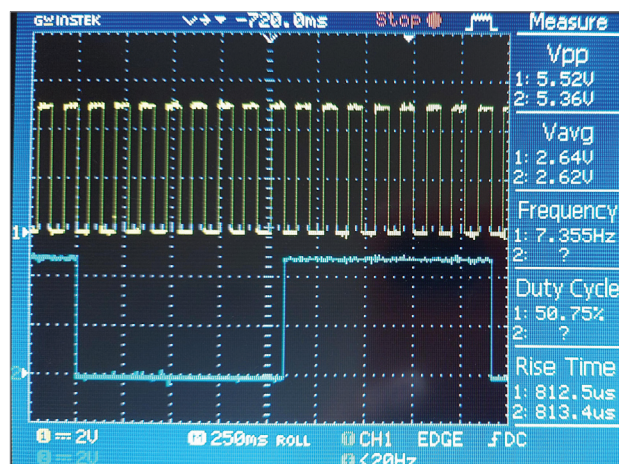


Fig. 3: Oscillator waveforms at pin 11 (yellow) and pin 7 (blue)